AIR POLLUTION AND CONTROL

Subject Code : 23A01504b

UNIT - II : METEOROLOGY AND AIR POLLUTION

Syllabus: Factors influencing air pollution, Wind rose, Mixing depths, Lapse rates and dispersion - Atmospheric stability, Plume rise and dispersion, Prediction of air quality, Box model - Gaussian model - Dispersion coefficient - Application of tall chimney for pollutant dispersion.

1. Introduction

Once pollutants are emitted they are carried, diluted and transformed in the atmosphere. **Meteorology** (study of atmospheric processes) decides how far and how fast the pollutants are dispersed. The same emission can cause serious pollution on one day and negligible pollution on the next day, only because of the weather.

- Atmosphere layers: Troposphere (0-12 km, weather, pollution) | Stratosphere (12-50 km, ozone) | Mesosphere | Thermosphere
- Most air pollution problems occur in the lowest part of the troposphere - the **planetary boundary layer** (up to 1-2 km)
- Dispersion = transport (advection by wind) + dilution (turbulent diffusion)

2. Factors Influencing Air Pollution

2.1 Meteorological factors

- **Wind speed:** concentration is **inversely proportional to wind speed** ($C \sim 1/u$). High speed -> more dilution

- **Wind direction:** decides which area is polluted (downwind); persistent direction gives high dose at a place
- **Temperature:** affects stability, rate of chemical reactions, plume buoyancy; temperature inversion traps pollutants
- **Solar radiation:** drives convection and mixing; also drives photochemical reactions (O_3, PAN)
- **Humidity and fog:** SO_2 converts to sulphuric acid mist; fog and low cloud increase pollutant retention
- **Precipitation:** washout (below-cloud scavenging) and rainout (in-cloud) clean the air; acid rain results
- **Atmospheric stability and turbulence:** most important - decide vertical mixing

2.2 Topographical factors

- **Valleys:** cold air drains down at night and pollutants collect in the valley (inversion trap)
- **Hills / buildings:** cause downwash, eddies, wake effects; plume impinges on hill slope
- **Land and sea breezes:** sea breeze by day, land breeze by night - can recirculate pollutants (coastal)
- **Mountain-valley winds:** anabatic (up-slope, day) and katabatic (down-slope, night)
- **Urban heat island:** warmer city centre, rising air, inflow of air near ground (dust dome)
- **Vegetation, water bodies:** change surface roughness and heating

2.3 Source and receptor factors

- Emission rate, stack height, exit velocity, exit temperature, location of the source
- Nature of pollutant: particle size, density, chemical reactivity, settling velocity
- Distance and elevation of the receptor from the source, population density

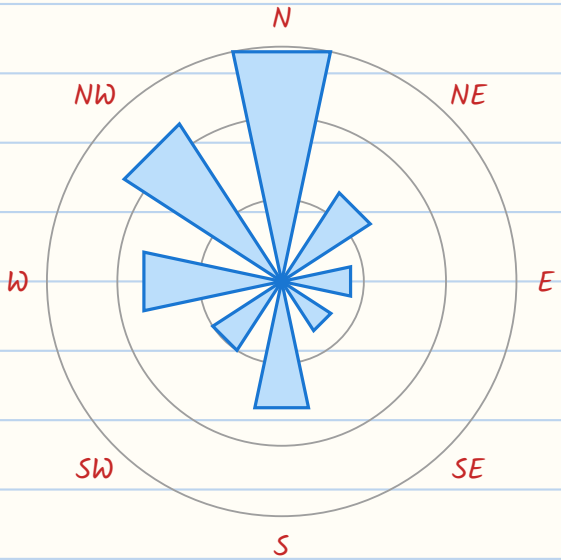
Pollution potential = Emission strength x Meteorological conditions x Topography

3. Wind and Wind Rose

Wind is horizontal movement of air due to pressure gradient. **Wind speed** is measured by cup **anemometer**, **direction** by wind vane (direction FROM which the wind blows). Speed increases with height: $u_2/u_1 = (z_2/z_1)^p$ where p = 0.1-0.4 depends on stability and roughness (power law).

Wind rose diagram

A **wind rose** is a graphical plot of the frequency of occurrence of wind from each direction (and speed class) at a location over a period (month, season, year).



petal length = % of time wind blows FROM that side
centre = calm %

Fig. Typical wind rose (16 or 8 points)

- The length of each spoke is proportional to the % of time the wind blows FROM that direction; the longest petal is the **prevailing wind direction**
- Different widths / shades on a spoke show wind speed classes; the circle at centre shows % calm (< 1 m/s)
- **Uses:** locating industries (on the downwind side of residential areas, with a green belt), siting air monitoring stations, planning airports and roads, choosing stack positions
- Example: if prevailing wind is from SW, the residential colony must be placed to the SW of the factory (upwind).

4. Lapse Rates

Lapse rate is the rate of decrease of air temperature with increase of altitude (dT/dz). Normally temperature falls in the troposphere.

Term	Meaning	Value
ELR (environmental / ambient lapse rate)	Actual temperature change of the surrounding air	Varies; mean 6.5 deg C/km
DALR (dry adiabatic lapse rate)	Cooling of a rising dry parcel (no heat exchange)	9.8 deg C/km (about 1 deg C/100 m)
SALR (saturated adiabatic)	Cooling of a rising saturated parcel (latent heat released)	4-6 deg C/km
Inversion	Temperature increases with height (ELR negative)	dT/dz > 0

A rising air parcel expands, does work and cools adiabatically: $T_2 / T_1 = (P_2 / P_1)^{0.286}$. Its motion depends on how its temperature compares with the surrounding air - hence stability.

Stability conditions

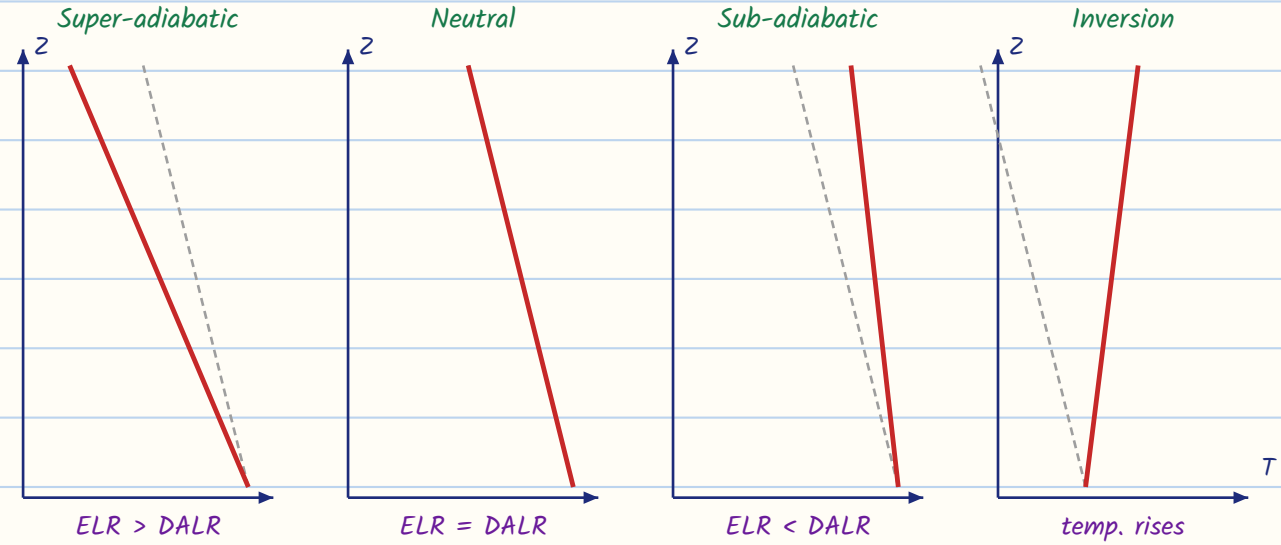


Fig. ELR (red) compared with DALR (dashed grey)

Condition	ELR vs DALR	Behaviour of parcel	Dispersion
Unstable (super-adiabatic)	ELR > 9.8	Displaced parcel keeps rising (warmer than surroundings)	Very good vertical mixing
Neutral	ELR = 9.8	Parcel stays where it is placed	Moderate
Stable (sub-adiabatic)	ELR < 9.8	Parcel returns to original level	Poor, suppressed mixing
Inversion	Temperature increases	Strongly stable lid	Very poor, pollutants trapped

Temperature inversions

- **Radiation (surface) inversion:** clear calm night, ground cools by radiation and cools the air above; breaks up after sunrise. Common in winter - morning smog
- **Subsidence inversion:** large-scale sinking of air in high-pressure region (anticyclone); air warms by compression; at 500-1500 m; can last for

days (Los Angeles)

- **Frontal inversion:** warm air overrides a cold air mass at a front
- **Advection inversion:** warm air moves over a cold surface (sea breeze over cold sea / snow)
- **Valley inversion:** cold air drains to the valley floor at night

Inversion = lid on the atmosphere. Pollutants cannot rise above it -> high ground-level concentration (e.g. Donora 1948, London 1952, Bhopal 1984).

5. Mixing Depth (Mixing Height)

Mixing depth (MD) is the height above the ground up to which vertical mixing of pollutants takes place by thermal turbulence. Above this height air is stable and acts as a lid.

- Determined graphically: plot the temperature-height profile (radiosonde) of early morning and draw a **dry adiabat** from the **maximum surface temperature of the afternoon**; the height where the dry adiabat meets the morning profile is the **maximum mixing depth**
- Typical values: afternoon 1000-3000 m (summer), 300-1000 m (winter), night / early morning 50-300 m
- **Ventilation coefficient (VC) = Mixing depth x mean wind speed in the mixed layer** (m^2/s)
- VC < 6000 m^2/s -> poor dispersion (high pollution potential); VC > 10,000 m^2/s -> good dispersion

Atmospheric stability classes (Pasquill-Gifford) - A to F:

Surface wind (m/s)	Strong	Moderate	Slight	Night >= 4/8 cloud	Night <= 3/8 cloud
< 2	A	A-B	B	-	-

Surface wind (m/s)	Strong	Moderate	Slight	Night >= 4/8 cloud	Night <= 3/8 cloud
2 - 3	A-B	B	C	E	F
3 - 5	B	B-C	C	D	E
5 - 6	C	C-D	D	D	D
> 6	C	D	D	D	D

A = extremely unstable, B = moderately unstable, C = slightly unstable, **D = neutral**, E = slightly stable, F = moderately stable (columns under 'strong/moderate/slight' are day-time solar insolation; D is used for overcast day or night).

6. Plume Behaviour (Types of Plumes)

The shape of the plume from a chimney depends on the vertical temperature profile (stability).

Plume type	Atmospheric condition	Features
Looping	Strongly unstable (super-adiabatic)	Large eddies, wavy; high conc. near stack at times; good dilution
Coning	Neutral (about adiabatic), cloudy and windy	Cone shape; spreads at equal rate up and down
Fanning	Strong inversion (stable) from ground to above stack	Spreads horizontally only, travels far, no vertical mixing
Lofting	Inversion below, unstable above the stack	Plume goes upward - favourable, no ground effect
Fumigation	Unstable below, inversion above stack	Pollutants brought to the ground - worst case, high GLC
Trapping	Inversion above and below stack height	Plume confined between two layers; ground level high

Best: lofting. Worst: fumigation (e.g. at sunrise when the nocturnal inversion breaks up) and trapping.

7. Plume Rise

The **effective stack height H = h_s + Dh**, where h_s = physical stack height and Dh = plume rise above the stack top. Plume rise occurs because of (i) **momentum** of the exit gases (velocity) and (ii) **buoyancy** (hot gases lighter than air).

Plume rise increases with: higher exit velocity V_s, higher stack diameter d, larger gas temperature difference (T_s - T_a), lower wind speed u and neutral / unstable atmosphere.

Holland's formula

- $Dh = (V_s\, d / u)\, [\, 1.5 + 2.68 \times 10^{-3}\, P\, d\, (T_s - T_a) / T_s\,]$
- V_s = stack exit velocity (m/s), d = inside diameter of stack tip (m), u = wind speed at stack height (m/s), P = atmospheric pressure (millibar), T_s = stack gas temp (K), T_a = air temp (K)
- For stable conditions reduce by 10-20%, for unstable increase by 10-20% (Holland's correction)

Briggs' formula (buoyancy-dominated)

- Buoyancy flux: $F = g\, V_s\, d^2\, (T_s - T_a) / (4\, T_s)$ (m^4/s^3)
- Final rise (neutral/unstable): for F < 55: $Dh = 21.425\, F^{0.75} / u$; for F >= 55: $Dh = 38.71\, F^{0.6} / u$
- Gradual rise at distance x: $Dh = 1.6\, F^{1/3}\, x^{2/3} / u$

Other formulae

- Concawe, Davidson-Bryant, Moses-Carson formulae are other empirical relations used for different conditions
- Plume rise is usually **underestimated** in calm / stable air and overestimated in strong winds - use judgement

Numerical idea: V_s = 15 m/s, d = 3 m, u = 5 m/s, T_s = 400 K, T_a = 300 K, P = 1000 mb. Dh = (15 x 3 / 5)[1.5 + 2.68e-3 x 1000 x 3 x 100/400] = 9 x [1.5 + 2.01] = **31.6 m**.

8. Prediction of Air Quality - Dispersion Models

Air quality models are mathematical tools to predict the concentration of pollutants at receptors for known emission and meteorology. Uses: planning (siting), impact assessment (EIA), design of stack height, control strategy, forecasting.

- **Gaussian plume models** (point source, steady state) - ISC3, AERMOD, CALPUFF
- **Box models** (area source), **Eulerian** grid models, **Lagrangian** trajectory models
- **Statistical / receptor models** (CMB) and physical (wind tunnel) models

8.1 Box model

Simplest model for an **urban area source**. The city is treated as a box of length L (along wind), width W, with a lid at mixing height H.

- **Assumptions:** (1) uniform emission over the area (2) pollutants are completely and uniformly mixed within the box up to H (3) wind direction constant, wind speed u uniform (4) **steady state** (5) pollutants are conservative - no chemical reaction or removal, (6) no

pollutant crosses the top lid

- Mass balance (inflow + emission = outflow): C_b u W H + q L W = C u W H
- **$C = C_b + q\ L / (u\ H)$** where q = emission rate per unit area ($g/m^2.s$), C_b = background (upwind) conc.

Example: q = 2 x 10^{-5} $g/m^2.s$, L = 10 km, u = 4 m/s, H = 500 m -> C = 2e-5 x 10000 / (4 x 500) = 1 x 10^{-4} g/m^3 = **100 ug/m^3** (above background).

Box model: C ~ L / (u H). Increases with larger city size and decreases with higher wind and deeper mixing. Poor for local (short distance) predictions.

8.2 Gaussian plume model

Most widely used model for **continuous point sources** (chimney). Assumes the concentration distribution in the crosswind (y) and vertical (z) directions is **Gaussian (normal)** with standard deviations $sigma_y$ and $sigma_z$. The origin is at the ground below the stack, x is downwind.

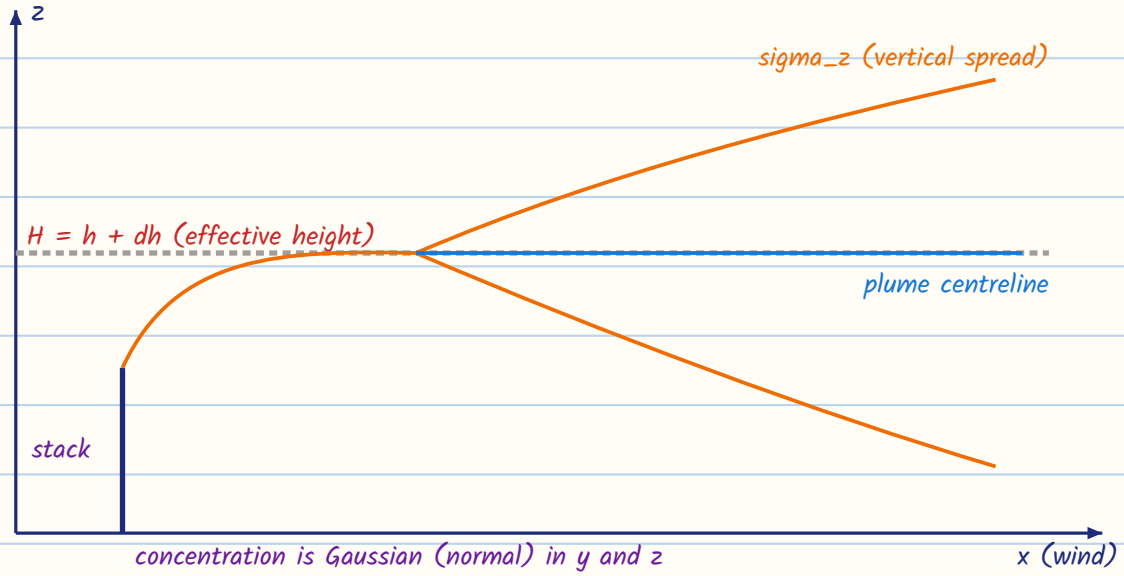


Fig. Gaussian plume - coordinate system and effective stack height

Assumptions: continuous steady emission; uniform wind u along x (constant with height); conservative pollutant (no reaction, decay or deposition); total reflection at ground (no absorption); flat open terrain; turbulent diffusion in x direction is negligible compared to advection; σ_y, σ_z depend only on x and stability.

$$C(x,y,z) = Q / (2\pi u \sigma_y \sigma_z) \times \exp(-y^2 / 2\sigma_y^2) \times [\exp(-(z-H)^2 / 2\sigma_z^2) + \exp(-(z+H)^2 / 2\sigma_z^2)]$$

- C = concentration (g/m^3), Q = emission rate (g/s), u = mean wind speed at stack height (m/s), H = effective stack height, σ_y, σ_z = dispersion coefficients (m)
- The second exponential term is the **image source** that accounts for ground reflection

Special cases

- **Ground level concentration (z = 0):** $C(x,y,0) = Q / (\pi u \sigma_y \sigma_z) \exp[-y^2/2\sigma_y^2 - H^2/2\sigma_z^2]$
- **Ground level centre line (y = 0, z = 0):** $C(x,0,0) = Q / (\pi u \sigma_y \sigma_z) \exp[-H^2/2\sigma_z^2]$
- **Ground-level source (H = 0):** $C(x,0,0) = Q / (\pi u \sigma_y \sigma_z)$
- **Maximum ground level concentration:** occurs where **$\sigma_z = H / \sqrt{2}$**, and $C_{max} = 2Q / (e\pi u H^2) \times (\sigma_z / \sigma_y) \sim 0.234 Q / (u H^2)$ when $\sigma_y = \sigma_z$
- With a **mixing lid**, multiple reflections between ground and lid are added when $\sigma_z > 0.8 L$ (beyond that distance the plume is uniform in z: $C = Q / (\sqrt{2\pi}\, u \sigma_y L) \exp(-y^2/2\sigma_y^2)$)

C_{max} is inversely proportional to H^2: doubling the effective stack height reduces maximum GLC to about 1/4. (and u: $C \sim 1/u$)

8.3 Dispersion coefficients (σ_y, σ_z)

Measure of the plume spread; they **increase with downwind distance** and **with instability**. Standard sources:

- **Pasquill-Gifford (Turner) curves** - σ_y and σ_z vs x for stability classes A-F (x from 0.1 to 100 km)
- **Briggs' formulae** (open country): class A: $\sigma_y = 0.22\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.20\ x$; class D: $\sigma_y = 0.08\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.06\ x\ (1 + 0.0015\ x)^{-1/2}$; class F: $\sigma_y = 0.04\ x\ (1 + 0.0001\ x)^{-1/2}$, $\sigma_z = 0.016\ x\ (1 + 0.0003\ x)^{-1}$ (x in metres)
- **Power law fit:** $\sigma_y = a\ x^{0.894}$; $\sigma_z = c\ x^{d} + f$ (Martin 1976) with constants a, c, d, f depending on class and x (km)
- Urban conditions have more roughness -> larger sigma (McElroy-Pooler, Briggs urban formulae)

Procedure to solve a problem: (1) find stability class from table (2) find effective stack height $H = h_s + \Delta h$ (3) read σ_y, σ_z at the required x (4) substitute in the Gaussian equation (5) convert g/m^3 to ug/m^3 ($\times 10^6$).

Example: Q = 100 g/s, u = 4 m/s, H = 60 m, at x = 500 m (class C, Briggs): $\sigma_y = 54$ m, $\sigma_z = 38$ m. $C(500,0,0) = 100 / (3.14 \times 4 \times 54 \times 38) \times \exp(-3600 / (2 \times 1444)) = 0.00388 \times \exp(-1.25) = 0.00388 \times 0.287 = 1.1 \times 10^{-3}\ g/m^3$ = **about 1100 ug/m^3**.

9. Application of Tall Chimney for Pollutant Dispersion

Instead of treating gases, the cheapest way historically has been the **'dilution' by tall stack**: release the pollutants at a high level so that they are dispersed over a very large area and the ground-level concentration (GLC) is low.

- GLC ~ $1/H^2$: a taller chimney drastically reduces the **maximum GLC** and shifts the point of maximum farther downwind
- Takes the plume above the **inversion/ building wake zone** and avoids **downwash** (aerodynamic) and **fumigation**
- **Stack height = h_s.** Effective height $H = h_s + Dh$; exit velocity should be at least **1.5 times** wind speed to prevent downwash; good practice: stack height >= **2.5 times** height of nearest building
- **CPCB stack height guidelines:** $H = 14\ Q^{0.3}$ (H in m, Q = SO_2 emission in kg/h) for power plants; thermal plants of >= 500 MW need 275 m; 200 - 500 MW 220 m; < 200 MW $14\ Q^{0.3}$
- Other industries: minimum stack height about 30 m as per State Pollution Control Board norms; IS 6533 gives the structural design of chimneys

Merits and demerits

Merits	Demerits
Reduces local GLC and nuisance	Does not reduce total emission; only transfers pollution
Low capital cost vs control equipment	Pollutants travel far - acid rain, transboundary effects
Simple operation, no energy penalty	Very high structure cost, aviation hazard, may still fumigate
Takes plume above inversion in many cases	Not acceptable as sole control under present standards

Modern rule: Use tall stacks AFTER installing proper control equipment, not instead of it. 'Dilution is NOT the solution to pollution'.

10. Quick Revision / Important Questions

1. Explain the factors influencing air pollution (meteorological and topographical).
2. What is a wind rose? How is it used in industrial siting?
3. Define lapse rate; explain ELR, DALR and the conditions of atmospheric stability.
4. Explain different types of plumes with neat sketches.
5. What is mixing depth? Define ventilation coefficient.
6. Derive the Gaussian plume equation; list assumptions. Derive C_{max} and its location.
7. Explain the box model with assumptions. Write notes on dispersion coefficients.
8. Discuss the use and limitations of tall stacks. Explain Holland's and Briggs' plume rise formulae.